
Samuel DalCais

Department of Chemistry | Colorado State University | Fort Collins, CO 80523

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EDUCATION

- 2021-2023 **Colorado State University**, Fort Collins, CO
Degree Expected: M.S. Organic Chemistry
- 2017-2021 **Wells College**, Aurora, NY
Degree Awarded: B.A. Chemistry (Junior/Senior GPA: 3.87)
Research Advisor: Prof. Lindsay Burwell
Henry Wells Scholarship Recipient

RESEARCH EXPERIENCE

Organic Chemistry: Developed phosphorous ligand coupling reactions for the regioselective functionalization of N-heterocycles under the supervision of Prof. Andrew McNally at Colorado State University.

Biochemistry: Studied the role of succinyl-CoA synthetase mutations in Leigh's Syndrome under the guidance of Dr. Lindsay Burwell — worked on replacing the succinyl-CoA synthetase-encoding gene in yeast with the human version so that the Leigh's Syndrome-associated point mutations in said gene could be studied *in vitro*.

SKILLS AND TRAINING

- ♦ Expertise in the synthesis and purification of organic compounds using conventional and inert atmosphere techniques.
- ♦ Skilled in the structural elucidation of organic molecules (MS, IR, GC, UV-vis, ^1H , ^{19}F , and ^{31}P NMR)
- ♦ Experienced in modern synthetic methods including organo-, transition metal-, and photoredox catalysis.
- ♦ Skilled in biochemical techniques such as protein extraction, SDS-PAGE, Western blotting, gel electrophoresis, PCR, and primer design.
- ♦ Basic Python programming skills.
- ♦ Effective communicator of results in both written form and oral presentations.

RELEVANT COURSE WORK

Undergraduate Courses:

General Chemistry
Organic Chemistry I & II
Biochemistry
Biology

Biochemical Techniques Lab
Inorganic Chemistry
Solid State Chemistry
Instrumental Analysis

Physical Chemistry
Physics I
Calculus I & II

Graduate Courses:

Synthetic Organic Chemistry I
Organic Molecular Structure Determination
Organometallics in Synthesis
Structure/Mechanisms in Organic Chemistry

Foundations of Inorganic Synthesis
Inorganic Chemistry: Group Theory
Inorganic Chemistry: Magnetic Spectroscopies

TEACHING EXPERIENCE

General Chemistry Teaching Assistant, CSU Chemistry
Organic Chemistry Teaching Assistant, CSU Chemistry

Fall 2021
Spring – Fall 2022

During my three semesters as a teaching assistant, I taught two laboratory classes per week, comprising 22 students a piece, wherein I gave lectures, helped undergraduates conduct experiments, and graded lab reports.

LANGUAGES

English: Native speaker
Hebrew: Reads fluently, speaks proficiently
Latin: Reads proficiently

REFEREES

Prof. Lindsay Burwell, Associate Professor of Chemistry - *Wells College*

Email: lburwell@wells.edu

Phone: (315)364-3268

Prof. Melissa Reynolds, Professor of Chemistry – *Colorado State University*

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